



LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written examination

The use of electronic devices and additional material (books, notes, etc.) is not permitted during the exam. You may freely utilise all results proven in the lecture. Justify clearly your answers and please write your anonymous code at the top of each sheet.

1. (a) Give an example of a group G and a G -set X such that X has exactly one G -orbit of cardinality p for each prime number p . (2 pts.)

Solution: Let $G = \mathbb{Z}$ be the additive group of integer numbers. For each prime number p , the set of cosets $\mathbb{Z}/p\mathbb{Z}$ has a transitive action of the group $\mathbb{Z}$ by left translation (Example 2.19) and hence it consist of a single orbit of cardinality p . The disjoint union

$$X = \coprod_{p: \text{prime}} \mathbb{Z}/p\mathbb{Z}$$

with the induced action of $\mathbb{Z}$ has the required property.

- (b) Let p be a prime number and G a group of order p^2 . Prove that G is abelian. (3 pts.)

Solution: Since G is a finite p -group, its centre $Z(G) \triangleleft G$ is a non-trivial (Corollary 4.5) normal subgroup; moreover, by Lagrange's Theorem, the centre $Z(G)$ must be a group of order p or p^2 . Hence, by Cauchy's Theorem (Theorem 4.3), there exists an element $a \in Z(G)$ of order p . Applying Lagrange's Theorem one more time, we see that the quotient group $G/\langle a \rangle$ by the normal subgroup $\langle a \rangle \triangleleft G$ is cyclic of order p with generator $b\langle a \rangle$, say. Hence, for $g, h \in G$ there exists integers i, j, k, ℓ such that $g = b^j a^i$ and $h = b^\ell a^k$; therefore

$$gh = (b^j a^i)(b^\ell a^k) = (b^j b^\ell)(a^i a^k) = (a^i a^k)(b^j b^\ell) = (a^k a^i)(b^\ell b^j) = (b^\ell a^k)(b^j a^i) = hg.$$

This shows that G is an abelian group.

2. Determine all the Sylow subgroups of the symmetric group $\mathfrak{S}_3$. (5 pts.)

Solution: The symmetric group $\mathfrak{S}_3$ has order $6 = 2 \cdot 3$ and therefore its non-trivial Sylow subgroups have order 2 or 3 by the First Sylow Theorem (Theorem 4.9). By the Third Sylow Theorem (Theorem 4.12), the number n_3 of 3-Sylow subgroups $\mathfrak{S}_3$ is of the form $n_3 = 3k + 1$ and divides 2; Hence $n_3 = 1$ and $\mathfrak{S}_3$ admits a unique Sylow 3-subgroup, namely the subgroup

$$\{e, (321), (231)\}.$$

Similarly, again by the Third Sylow Theorem (Theorem 4.12), the number n_2 of Sylow 2-groups of $\mathfrak{S}_3$ is of the form $2k + 1$ and divides 3; hence $n_2 = 1$ or $n_2 = 3$. However, $\mathfrak{S}_3$ contains 3 subgroups of order 2 (necessarily Sylow 2-subgroups by Corollary 4.10), namely the subgroups

$$\{e, (21)\}, \quad \{e, (32)\}, \quad \text{and} \quad \{e, (31)\}.$$

3. Let R be a ring and suppose given a commutative square of R -module homomorphisms (indicated with solid arrows below) (5 pts.)

$$\begin{array}{ccccc} M & \xrightarrow{f} & N & \xrightarrow{p} & \text{coker } f \\ \downarrow u & & \downarrow v & & \downarrow \alpha \\ M' & \xrightarrow{g} & N' & \xrightarrow{q} & \text{coker } g \end{array} \quad v \circ f = g \circ u$$

where p and q are the canonical maps to the respective cokernels and u and v are isomorphisms of R -modules.

Prove that there exists a unique R -module homomorphism $\alpha: \text{coker } f \rightarrow \text{coker } g$ rendering the above diagram commutative (indicated with a dashed arrow above) and, moreover, that α is an isomorphism of R -modules.

Solution: The universal property of $\text{coker } f$ (Theorem 6.39) applied to the composite

$$(q \circ v) \circ f = q \circ (g \circ u) = q \circ 0 = 0$$

yields the required R -module homomorphism $\alpha: \text{coker } f \rightarrow \text{coker } g$. To prove that α is an isomorphism we shall construct its inverse. For this, notice that

$$v \circ f = g \circ u \implies f \circ u^{-1} = v^{-1} \circ g.$$

Hence, by the apparent variant of the previous argument we obtain a commutative diagram of R -module homomorphisms

$$\begin{array}{ccccc} M' & \xrightarrow{g} & N' & \xrightarrow{q} & \text{coker } g \\ \downarrow u^{-1} & & \downarrow v^{-1} & & \downarrow \beta \\ M & \xrightarrow{f} & N & \xrightarrow{p} & \text{coker } f \end{array}$$

We claim that $\beta = \alpha^{-1}$. Indeed,

$$(\beta \circ \alpha) \circ p = \beta \circ (q \circ v) = (p \circ v^{-1}) \circ v = p = id_{\text{coker } f} \circ p.$$

Since p is surjective, we have $\beta \circ \alpha = id_{\text{coker } f}$. A similar argument shows that $\alpha \circ \beta = id_{\text{coker } g}$ so that $\beta = \alpha^{-1}$, as required.

4. Let $\mathbf{k}$ be a field and $R := \text{Mat}_{2 \times 2}(\mathbf{k})$ the ring of 2×2 matrices with entries in $\mathbf{k}$. Fix $\lambda \in \mathbf{k}$ and consider the following subsets of the ring R : (5 pts.)

$$P := \left\{ \begin{pmatrix} a & -\lambda a \\ c & -\lambda c \end{pmatrix} \in R \mid a, c \in \mathbf{k} \right\}, \quad Q := \left\{ \begin{pmatrix} 0 & b \\ 0 & d \end{pmatrix} \in R \mid b, d \in \mathbf{k} \right\}.$$

Prove that P and Q are projective R -modules.

Solution: The matrices

$$\begin{pmatrix} 1 & -\lambda \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & \lambda \\ 0 & 1 \end{pmatrix}$$

form a complete set of pairwise orthogonal idempotents (Definition 6.48) in the ring R . We see at once that

$$P = R \cdot \begin{pmatrix} 1 & -\lambda \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad Q = R \cdot \begin{pmatrix} 0 & \lambda \\ 0 & 1 \end{pmatrix}$$

and that there is an isomorphism of R -modules (Theorem 6.49)

$$R \cong R \cdot \begin{pmatrix} 1 & -\lambda \\ 0 & 0 \end{pmatrix} \oplus R \cdot \begin{pmatrix} 0 & \lambda \\ 0 & 1 \end{pmatrix} = P \oplus Q.$$

Therefore P and Q are direct summands of a free R -module of rank one and hence projective R -modules (Theorem 7.34).

5. (a) Let R be a ring and $M = S_1 \oplus S_2 \oplus \dots \oplus S_n$ a semisimple R -module. Prove or give a counter-example to the following statement: (2 pts.)

The semisimple R -module M admits only finitely many R -submodules.

Solution: The statement is false. Consider the field $\mathbb{R}$ of real numbers and the $\mathbb{R}$ -vector space $\mathbb{R}^2$, which is the direct sum of two simple $\mathbb{R}$ -vector spaces. We see that $\mathbb{R}^2$ admits infinitely many one-dimensional $\mathbb{R}$ -subspaces (one for each line through the origin).

- (b) Consider the ring $\mathbb{R}[X]$ of univariate polynomials with coefficients in the field $\mathbb{R}$ of real numbers. Prove that the $\mathbb{R}[X]$ -module S determined by the pair $(\mathbb{R}^2, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix})$ is simple. (3 pts.)

Solution: We claim that S is isomorphic to the $\mathbb{R}[X]$ -module $\mathbb{R}[X]/(X^2 + 1)$, which is simple since the polynomial $X^2 + 1$ is irreducible in $\mathbb{R}[X]$ (see the Correspondence Theorem 6.52). Indeed,

$$\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{R} \cdot 1 \oplus \mathbb{R} \cdot X$$

as $\mathbb{R}$ -vector space and the matrix of the $\mathbb{R}$ -linear map $v \mapsto X \cdot v$ with respect to the ordered basis $(1, X)$ is clearly the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

6. Determine the elementary divisors and the invariant factors of the abelian group (5 pts.)

$$(\mathbb{Z}/30\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}/10\mathbb{Z}.$$

Solution: Recall the isomorphism of abelian groups (Corollary 11.14)

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/(m\mathbb{Z} + n\mathbb{Z}) = \mathbb{Z}/\gcd(m, n)\mathbb{Z}, \quad m, n \geq 1.$$

Since tensor products distribute over direct sums (Corollary 11.12), we obtain the following isomorphisms of abelian groups

$$\begin{aligned} (\mathbb{Z}/30\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}/10\mathbb{Z} &\cong (\mathbb{Z}/30\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/10\mathbb{Z}) \oplus (\mathbb{Z}/5\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/10\mathbb{Z}) \\ &\cong \mathbb{Z}/10\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z} \\ &\cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z} \oplus \mathbb{Z}/5\mathbb{Z}. \end{aligned}$$

We see that 10, 5 are the required invariant factors while the elementary divisors are 2, 5, 5.